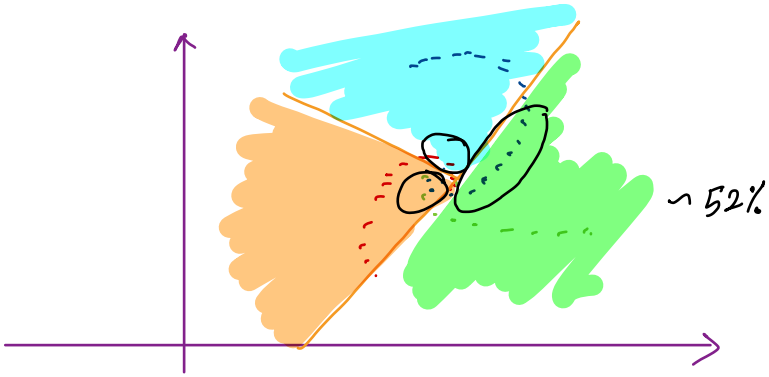
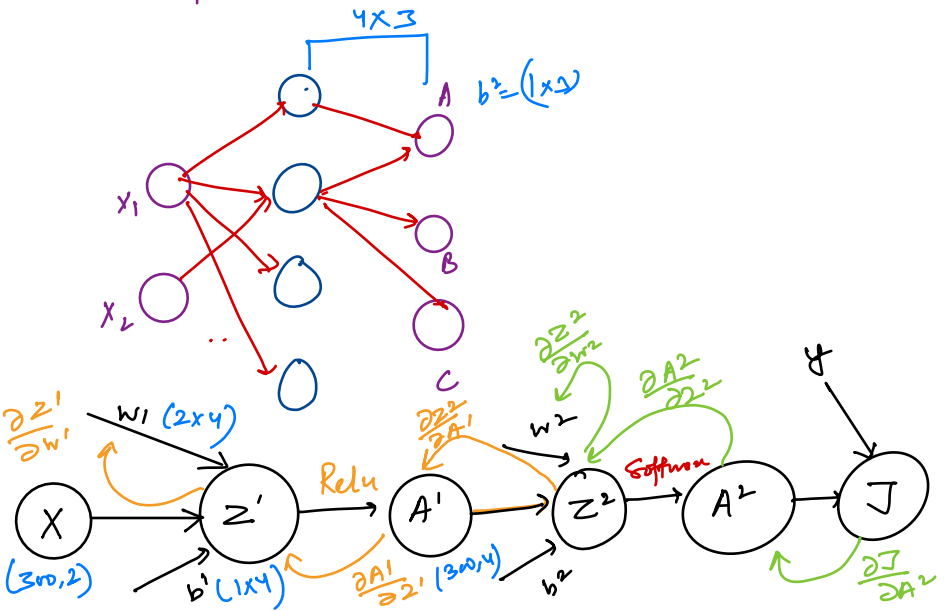
Agenda

1. Backpropogation
2. Calculating Gradients
3. Summarizing the process

} Scratch

4. Train a NN using Sklearn → 2 line implementation





$(3\omega \times 2) \times (2 \times 4)$ (1×4)

$Z^1 = np.dot(X, W_1) + b_1$

$A^1 = Relu(Z^1)$ $(3\omega \times 4)$

$\Rightarrow np.max(0, Z_1)$

$Z^2 = A^1 W^2 + b^2$

$\Rightarrow (3\omega \times 4)(4 \times 3) + (1 \times 3)$

$\Rightarrow (3\omega \times 3)$

$A^2 = (3\omega \times 3)$

$$W^{1}_{new} = W^{1}_{old} - \alpha \frac{\partial J}{\partial W^1}$$

$$W^{2}_{new} = W^{2}_{old} - \alpha \frac{\partial J}{\partial W^2}$$

$$\frac{\partial J}{\partial W^2} = \frac{\partial J}{\partial A^2} \times \frac{\partial A^2}{\partial Z^2} \times \frac{\partial Z^2}{\partial W^2}$$

$$\frac{\partial J}{\partial W^1} = \frac{\partial J}{\partial A^2} \times \frac{\partial A^2}{\partial Z^2} \times \frac{\partial Z^2}{\partial A^1} \times \frac{\partial A^1}{\partial Z^1} \times \frac{\partial Z^1}{\partial W^1}$$

(1)

$(a - y)$

$dZ_2 = A^2 - y$

$\downarrow$

probs $-$ y → one-hot encoded

$(3\omega, 3)$ $(3\omega \times 1)$

$(3\omega \times 3) - (3\omega \times 1)$

$\Rightarrow (3\omega \times 3)$

$$dW_2 = \underbrace{(A^2 - y)}_{dz_2} \times \underbrace{A'}_{\frac{\partial z^2}{\partial w^2}}$$

$(3m \times 3) \qquad (3m \times 4)$

$A'.T \qquad (A^2 - y)$

$(4 \times 3m) \qquad (3m \times 3)$

(4×3)

$$db_2 = \frac{\partial J}{\partial A^2} \times \frac{\partial A^2}{\partial z^2} \times \frac{\partial z^2}{\partial b^2}$$

$z^2 = w^2 A^1 + b^2$

$\frac{\partial z^2}{\partial b^2} \Rightarrow 1$

$db_2 = (A^2 - y) \times 1$

$(1 \times 3) \qquad (3m \times 3) \times 1$

$db_2 = dz^2$

$(1 \times 3) \qquad (3m \times 3)$

sum $(1 \times 3) \leftarrow$

mean (1×3)

$$\frac{\partial J}{\partial W^1} = \frac{\partial J}{\partial A^2} \times \frac{\partial A^2}{\partial Z^2} \times \frac{\partial Z^2}{\partial A^1} \times \frac{\partial A^1}{\partial Z^1} \times \frac{\partial Z^1}{\partial W^1}$$

dA^1 — $[0,1]$ — dz_2

$Z^2 = W^2 A^1 + b^2$

$\frac{\partial Z^2}{\partial A^1} = W^2$

$dA^1 = dz_2 \times W^2$

$A^1 = Relu(Z^1)$

$= \max(0, Z^1)$

$\frac{\partial A^1}{\partial Z^1} \Rightarrow 1 \text{ or } 0$

$dz_1 = dA_1 \times [0 \text{ or } 1]$

Relu

$Z > 0 \rightarrow Z$

$Z \leq 0 \rightarrow 0$

$$\frac{\partial A^1}{\partial Z^1} = \frac{\partial Relu(Z^1)}{\partial Z^1} = \frac{\partial(0, Z_1)}{\partial Z_1}$$

$\rightarrow 0 \text{ or } 1$

$$dw_1 = dz_1 \times \frac{\partial(x w_1 + b_1)}{\partial w_1}$$

$dw_1 = dz_1 \times x$